$$\mathrm{pr}_2((1)) = \begin{bmatrix} \mathbf{A}_g & \mathbf{0} \\ \mathbf{B}_f \begin{bmatrix} \mathbf{C}_g \\ \mathbf{0} \end{bmatrix} & \mathbf{A}_f \end{bmatrix} + \begin{bmatrix} \mathbf{B}_g & \mathbf{0} \\ \mathbf{B}_f^{[:,1:i]}\mathbf{D}_g & \mathbf{B}_f^{[:,i+1:i+k]} \end{bmatrix}^{[:,m+1:m+k]} \mathbf{F} \begin{bmatrix} \mathbf{D}_f^{[:,1:i]}\mathbf{C}_g & \mathbf{C}_f \end{bmatrix}^{[j+1:j+k,:]},$$